

Generation Interconnection System Impact Study Report

For Queue Project Q20-1628

Okawville 138 kV
December 28, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1628. The Interconnection Customer proposes a 211.3 MW wind facility interconnecting at the Okawville 138 kV substation in the Pennsylvania area. The request was received on December 22, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 387 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$11,847,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The W Gallatin 345-138 kV TX is overloaded to 111.3% of its 375 MVA emergency rating for the loss of the West Gallatin to Felida 345kV L. The project contributes 12.3% DFAX (25.9 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Gallatin 345-138 kV TX	Replace transformer with higher-rated unit	\$11,847,000	100% (\$11,847,000)
	Total allocated to project		\$11,847,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.